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MediAssist: An ML-Based Healthcare Support Tool for Diagnosis and Prescription

Project Report submitted in the partial fulfilment
of
Bachelor of Technology
in
Information Technology

Submitted To

SVKM's NMIMS,
Mukesh Patel School of Technology Management & Engineering,
Shirpur Campus (M.H.)

Submitted By:

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Under The Supervision Of:

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INFORMATION TECHNOLOGY DEPARTMENT
Mukesh Patel School of Technology Management & Engineering
ACADEMIC SESSION: 2025-26

CERTIFICATE

This is to certify that the project entitled MediAssist: An ML-Based Healthcare Support Tool for Diagnosis and Prescription has been done by

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under my guidance and supervision & has been submitted partial fulfilment of the degree of "**Bachelor of Technology** in **Information Technology"** of MPSTME, SVKM'S NMIMS (Deemed-to-be University), Shirpur (M.H.), India.

Prof. Atul Patil **Examiner**

(Internal Guide)

Date:

Place: Shirpur

H.O.D.

INFORMATION TECHNOLOGY DEPARTMENT

Mukesh Patel School of Technology Management & Engineering

ACKNOWLEDGEMENT

Honestly, I met numerous hardships in the entire process of this capstone project which were very tough to get through but on the other hand my only way to relieve the stress was focusing on the good things happening around me and that was good enough for me to reach my target; even above this feeling is my thankfulness to The Almighty for allowing me to experience this moment in life.

It gives us immense pleasure to express our sincere gratitude to **Prof. Atul Patil**, who is the **Associate Professor in the IT Department at MPSTME, Shirpur**, for his unwavering support and encouragement throughout my whole project, which made me very enthusiastic about my work. It is hard to imagine what my journey would have been without their constant support, understanding, and expert guidance. Their vast and profound knowledge in the IT field opened my eyes to the fact that just mastering the theory is only the first step in the marathon of creating effective operational software, that is, the integration of all core subjects of the field. I am really thankful to them for their mentoring which is priceless and support that is always coming my way, which altogether allowed me to finish this project right on time. Mentoring by them has been a fruitful and picture-perfect experience.

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We would like to express my sincere appreciation to all IT department personnel and to everyone that helped me comprehend technical computer technology at various stages of my B.Tech degree in Information Technology.

Talking about divisive, conflict-eliciting topics and issues can spur disagreement and acts against peoples' passionate beliefs.

The Capstone Project, fundamentally, is not an individual task but a collective team activity, where the end product is considered as a result of the teamwork done. The contribution of each person is recognized in the credits as important to the project's success. This section is primarily meant to show appreciation and convey acknowledgment to all the people who have, in their own distinct ways, directly or indirectly, contributed to the project up to this point and without whom the project would still be in the unfinished stage.

Grishma Khairnar
Vedashri Patil

ABSTRACT

The prompt and correct diagnosis of the disease still remains a major hurdle to the use of health care services in most of the places where there is a lack of medical personnel. The project called "MediAssist: An ML-Based Healthcare Support Tool for Diagnosis and Prescription" takes care of this problem by combining a strong web application that carries out automatic disease diagnosis based on the user's symptoms and also gives the necessary medical information using Computer Science ML-based techniques. The application is founded on a stacking model that incorporates Random Forest and XGBoost as the base learners and Logistic Regression as the meta-learner. It has been evaluated on a dataset comprising 41 different diseases and has attained an astonishing accuracy of 93.52% which demonstrates that the model can be very useful in predicting possible diseases. MediAssist is an easy-to-use application that initially collects the symptoms, then forecasts the disease that is most likely to occur and finally provides the user with information such as definition, moderation tips, and medicines prescribed. The product is meant to function as an additional health care system that will raise the awareness and therefore the early consultations, but it is not to replace diagnostics. Ethical concerns, data privacy, and transparency have been given the utmost importance throughout the development process like never before. The roll-out of the new system is a loud signal to the world of the growing power of AI-based solutions in health care, providing primary medical support that is accessible, efficient, and scalable.

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Chapter 1
Introduction

The process of prompting and accurately diagnosing diseases and the long and expensive medical procedures that are still the main obstacles of the whole scenario, have become extremely important not only for the patients' treatment but for the patients' lives. One of the most successful applications of artificial intelligence (AI) has been the introduction of machine learning (ML) into the medical field as a powerful weapon for unraveling the complexities of data and providing essential insights. MediAssist, the new web-based application developed through the project, is built on ML technology. The goal of this application is to support the public by providing them with information about possible diseases according to the symptoms they themselves have mentioned. The system uses a hybrid stacking model for predicting potential diseases and also provides additional information such as disease descriptions, preventive measures, and mainstay medications.

1.1 Background of the project topic

A quick and accurate diagnosis is the very feature of an ideal and even surpassing healthcare system of modern times. The diagnosis is the initial step in therapy, and thus, it is highly important as it directly affects the patient's condition. However, the global access to rapid diagnoses is still a major issue and the first to suffer from it are the poor in developing or remote areas.

Access to competent healers is invariably hampered by a variety of hard factors. What is the distance separating the patients from the doctors, the vast medical knowledge that the doctors may not even be able to completely cover themselves, and the consultation fee that is most of the time too high for the patients? People who are suffering from these above-mentioned money and logistics problems are, in some cases, not diagnosed on time which is mainly due to these factors. One of the consequences of such delays could be the wrong or late treatment allowing the disease to spread, which in turn, prolongs the patient's suffering, makes them a health cost incidence, and possibly results in the patient dying in the future due to the disease.

On the other hand, the Machine Learning (ML) technology is the one that is still the most powerful and the most effective among the new technologies applied in the healthcare industry. The power of ML algorithms is mainly built on their outstanding capability to detect even the most intricate data patterns and consequently offer very valuable and insightful data-based conclusions. This very power is perfect for the medical diagnosis challenge.

The use of machine learning for predicting diseases based on symptoms has been acknowledged, but mainly as a support tool rather than a substitute. The method is a very significant first step, makes the procedure easier, and is a provider of leading knowledge. The plan of the system is to receive the user's described symptoms, inform them about possible health issues, and most importantly, direct them to correct and timely medical treatment.

1.2 Motivation and scope of the report

The investigation into the use of machine learning in medical diagnostics has been exhaustive and has revealed both the advantages and the disadvantages of this technology, which is the main reason for the project. Disease forecasting, which is a challenging and highly risky area, needs the highest accuracy and reliability that conventional algorithms cannot provide most of the time.

- Limitations of Traditional Models: Contrary to being only a reference point for numerous new models, the traditional ones still face considerable barriers when it comes to processing information about complex symptoms.
  - "Curse of dimensionality" is the phrase denoting the drawback that greatly impacts the decision tree algorithms. A dataset comprising 132 symptoms (features) will make "proximity" very costly in terms of computation and very questionable from a statistical point of view.
  - Naïve Bayes and other similar models rely on a "naive" assumption that all conditions are independent - which means that they regard the symptoms of the disease occurring together as completely disconnected. However, this is a very coarse simplification in the medical domain because cough and fever, for example, are usually interrelated. Thus, the wrong hypothesis limits the accuracy of the model to a certain level.

- Hybrid Ensembles Taking the Lead: It cannot be denied that hybrid or ensemble techniques are very powerful and effective. Eventually, the present-day methods practically get rid of the disadvantages of the separate classifiers.
    - The addition of individual powerful algorithms collectively combines their strengths.
    - Union modeling would allow both models to eliminate each other's weaknesses. For example, the model that has a lot of variance (overfitting) would offset the model that has little variance. The final prediction merger produces a system that is overall more accurate and trustworthy than the sum of its parts.

The present paper presents the MediAssist application that aims to implement the most recent machine learning techniques to provide medical help that is not only very real but also quite accessible. Just the idea of having a system that would be more trustworthy than the current ones has been one of the greatest driving forces behind the project.

The undertaking of this project will be to completely transform the whole process starting from raw data to a user-friendly application framework by passing through all the processing steps, which is an enormous responsibility.

1. Application Framework: The anticipated application will be a novel, internet-based, and simple-to-use for everybody. Providing this great diagnostic tool to the end users through an attractive interface is the main objective; it would help to tackle the central problem of poor access to instant medical information.
2. Implementation of Hybrid Model: The hybrid stacking model, which was innovative, formed the technical support of MediAssist. The application area does not limit itself to a single classifier but encompasses a complex multi-layered ensemble:
    - Base Learners: Just these two classifiers - Random Forest (RF) and XGBoost, are selected, which will be blended particularly. RF (a bagging model) is an excellent choice for handling noisy data and it also has a very nice quality to avoid overfitting. XGBoost, the boosting algorithm that has become the best among algorithms, in contrast, is recognized for its capability to simply identify complex, non-linear relationships in data.

- Meta-Learner: Among all the contenders, the Logistic Regression model is regarded as the strongest meta-learner. Its task is to determine the most reliable single prediction that incorporates the "expert opinions" of the two models in the most appropriate manner concerning RF and XGBoost.

3. User Comprehension and Backing: The project has widened its horizon from just prediction to many applications. One of the foremost goals will be to foster user comprehension through the supply of immediate and relevant context. Hence, the system will not only predict a possible illness but also gather and display all the pertinent and indispensable information that consists of:
   - A concise explanation of the disorder is required for her.
   - List of things a user could actually work on.
   - Drugged, it is mainly lessened based on certain criteria much liked by many.
4. Ethical Boundary: Eventually, the separation is allowed as the technology is categorized as a primary healthcare support device. MediAssist can very well and indisputably be seen as an information tool that helps the users in their decision making and in the search for a qualified medical practitioner. So, it is not a substitute for medical diagnosis or consultation, which is the concealing of service that is inherent to system's design.

1.3 Problem statement

The main issue around which the entire project is built is the issue of long delays. This problem comes from the very first signs of a patient, that is the waiting time until a right medical diagnosis is made by the doctor and he/she starts the correct treatment. The situation is sometimes even worse in the poorly developed areas and those rural regions where a doctor might not be as high-quality as you would expect, the distance may be too long or the consultation fee may just be as high as the patient's lousy financial state can bear. This so-called "diagnosis gap", or whatever it is called, has been the very reason why treatment delays are considered critical and the most important factor in the occurrence of such delays. In some instances, these delays can not only worsen the disease but also become the source of the patient's distress. At the same time, the healthcare system's consumption for a longer duration is increased, which eventually results in higher overall healthcare costs.

Looking at things through the technology lens, the problem still exists as a two-sided issue. To start with, a vast number of machine learning models are currently being used for predicting diseases, however, the majority of them still rely on the traditional method of just one algorithm which is decision trees. Even if they are primitive models like this one, there is a very high chance that such sophistication and accuracy using hybrid or ensemble techniques which are the latest methods one will not be able to reach. Besides, they cannot help but lose one of the prime signals that indicate the right diagnosis which are the complex and non-linear interactions among different symptoms. To make things worse, the researchers' models very often output mere guesses rather than predictions. A case in point is when the disease's name is announced, it is not a serious issue for the patient. The problem is not only with disease prediction but also with the absence of actionable intelligence. The affected person is then left with questions like "What does this mean for me?" and "What should I do next?". The issue that the project is tackling is the absence of an integrated, precise, informative, and user-friendly healthcare support tool.

1.4 Salient contribution

The project has not only produced extraordinary results but also shown some very creative applications of machine learning in the medical sector. Such outcomes are not only relative among algorithms but also signify the establishment of a system that is robust, productive, and ready for use.

The clarification of methods through discussion, the invention of integrated stacking ensemble model for disease prediction, and the thorough evaluation of that model make up the main and most significant technical contribution. The constructed is a tremendous shifting deli that was created with one objective in mind— to get rid of the drawbacks inherent in using single isolation classifiers. The model cleverly utilizes the different strengths of the two powerful tree-based ensembles by selecting them as base learners: Random Forest (RF), which is highly resistant to noise, and XGBoost, which is best known for its ability to reveal very complex, non-linear patterns. Then, this structure uses a Logistic Regression model as a final, high-level meta-learner to integrate the predictions from both base models, thus resulting in a more accurate and powerful forecast.

One huge and powerful advantage was solely the hybrid model under discussion's performance level, which could be described as the best without difficulty. It was trained on an enormous data set composed of signs and symptoms of 41 common diseases and achieved an unbelievably high and dependable test accuracy of 93.52%. This result can be interpreted as a very strong affirmation for the whole hybrid approach indicating that the model is intelligent enough to deal with various scenarios, hence, it can be the reliable processor for a healthcare support tool in a very wide application. Besides the model reaching an incredibly high accuracy, it also got fantastic results when it comes to macro-average precision (0.9248) and recall (0.9138), indicating that the model's accuracy was quite consistent across the 41 disease categories.

Moreover, the project is boundless of prediction by not just offering access but also machine learning integration with an information retrieval system that is super easy and very useful. The collaboration is, in fact, bridging the theoretical models (which typically yield a prediction label) with a good and practical application at the same time. The MediAssist system displays the prediction as the primary information that the user needs most.

1.5 Organization of report

The document that has been produced represents the "MediAssist" project in its entirety as it provides an exhaustive description of its aims, techniques, and results.

The project's context is a dilemma, and the quick and uncomplicated healthcare diagnosis which is a very hard task indeed in areas with fewer resources is a main worry. The introduction not only presents the research idea but also the problem with the subheadings—mainly the provision of actionable, contextual information (e.g. precautions, medications) as well as a prognosis—and articulates the goals that the study seeks to achieve.

Following the introduction, the literature review provides a comprehensive discussion of the current technologies and methodologies in the field of symptom-based disease prediction. It does so by indicating related works from the classical machine learning to the recent hybrid ensemble strategies, identifying the shortcomings of the current practices (such as the lack of integrated, user-friendly support tools) and placing the new system's significance in a larger context.

The approach used to develop the system is explained thoroughly in this section of the paper. It clarifies the data collection and preprocessing steps for the four datasets (symptoms, descriptions, precautions, and medications) and the software interfaces (e.g., Scikit-learn, Pandas, XGBoost, and Joblib) that were used. Moreover, the various transformations and applications to the data, features, and models, especially the hybrid stacking model (Random Forest + XGBoost) construction and training, are all mentioned. As a result, this section means of communication to the interested parties the methodologies employed in the system's development. It also demonstrates the development process as being rigorous and strict.

The report, in the end, offers a discussion on the future possibilities of the system and its improvement, including, among other things, the larger dataset with more diseases, and the model's interpretability being enhanced. A full-scale and secure web application deployment of the system is also a long-term possibility. This aspect of foresight emphasizes the continuous.

The project was important and expected to cause a favorable impact on many areas regarding the improvement of health care access. The organization of the report allows the reader to follow the author through the complicated trails of the project's difficulties and to see the advantages of the research area alongside.

Chapter 2

Literature survey

2.1 Introduction to overall topic

In the medical sector, machine learning (ML) has been and is still the only groundbreaking technology that enables providing health services of the highest quality and that are friendly to patients through the deciphering of the complex data types and the provision of insights where patients get positively impacted [1]. Conversely, the broad spectrum of AI general modalities in the healthcare domain not only facilitates the distinctiveness but also the amalgamation of ideas, innovations, and risks. The use of AI in healthcare is always scrutinized through the lens of ethical concerns around the patient's rights, therefore making the situation of the need for an extensive description of the various utilizations and the gigantic issues that are still waiting to be solved even more acute [2].

Data is the key element in these applications, which is, in the case of the healthcare sector, of an extremely complicated, heterogeneous, and sensitive nature. This is not only the case for well-organized EMRs (Electronic Medical Records) but also for disorganized clinical notes, patient-reported outcomes, and real-time biometric data. The research literature acknowledges that one of the major issues lies not only with the algorithm but also with the preprocessing and feature engineering that transform this data into a usable form [3]. Furthermore, it is crucial to consider ethical dilemmas such as patient confidentiality and the data bias that existed in the past, which are not only very serious but also very tightly linked to the very basis of the responsible growth of these ML systems.

An important question is whether to build a so-called smart pre-diagnosis or disease prediction system [4]. Such systems make use of data from various sources; for instance, some models are confined to electronic medical records (EMRs) or structured diagnosis codes, while others - like the one in this project - are based on user-reported symptoms [5].Generally, the ultimate aim is to create a healthcare system capable of accurately predicting events and at the same time recommending the right medications or therapies.[6]. Notwithstanding, it is very difficult to implement such models in practice, especially when they must be formulated as user-friendly and trustworthy online applications for the healthcare industry. Many papers in the area indicate a comparative assessment of different algorithms and also raise the ethical issues and responsible design practices that are required.

2.2 Exhaustive literature survey

The use of machine learning in the medical field is a vast research area, and the most basic investigation is already uncovering its tremendous power along with certain limitations. Davenport and Kalakota [15] were the pioneers who identified the potential impacts of AI on healthcare and suggested the ways such systems could enhance triage accuracy and speed up the process. On the other hand, Rahmani et al. [13] opted for a more organized approach and the entire area of ML in medicine was scrutinized through an exhaustive literature review. They not only recognized the various uses of ML, but also pointed out the most critical difficulties that must be overcome such as data quality, model interpretability, and practical issues concerning clinical integration.

The creation of AI systems for the purpose of pre-diagnosis and disease prediction is the foremost field and at the same time the primary aim of the project stated above. One of the main issues in this field is the various means of getting such data, e.g. the patient's account of his/her symptoms, and it all funnels down to whether the entire process is feasible or not. Sood and Sharma [7] referred to the very first machine learning application in the area of diagnosing diseases from symptoms when they said that ML could reveal the symptoms' patterns that lead to certain diseases. This claim was supported by Liu and Luo [10] who introduced a smart pre-diagnosis system which was completely reliant on symptoms thus the assertion that this data stream could by itself be a feasible one without the aid of more complicated (and often less accessible) data sources like EMRs and lab results had been corroborated.

Once the symptoms-based prediction research activities had been validated, they then looked into the most efficient ML algorithms. Sirigineedi et al. [2] addressed this concern by proposing a machine learning method that not only evaluated the performance of the various classifiers on symptom data but also picked Random Forest as the best performer which was in line with a broader tendency in the literature. Similarly, Fu-Fier-Path et al. [6] also took the same route by concentrating on the creation of superior machine learning classifiers for symptom-based screening.

Their study concludes that relying solely on the "off-the-shelf" models according to the standard is not enough, and the selection of the model and hyperparameter tuning are vital for achieving the desired accuracy level in medical applications.

The use of machine learning was regarded as a challenging yet unavoidable move in the detection process of Alzheimer's disease, which was performed by Wallensten et al. [5] who managed to discover the underlying causes of deaths and the different medications used together. Similarly, Zhu et al. [9] proposed a prediagnosis system for one of the ailments (pediatric chronic cough) that heavily leaned on a CNN, that is, the Medical-Semantic aware Convolutional Neural Network (MSCNN), which is very much powerful. The previously mentioned experiments unmistakably indicate that the specialty has started to shift towards more pronounced and refined infrastructures.

Among the trends in the literature that move from artificial intelligence to integrated recommendation systems, this transition is significant and at the same time very meaningful to the "MediAssist" project. By conducting a user requirement study, the researchers have come to a point where they state that merely providing a disease label is not sufficient for an end-user of the type. Nayak et al. [4] were the first ones to say that, presenting a smart model that supports both disease prediction and preventing the drug from being prescribed. Rupadevi and Gopikrishna [12] talk to a similar extent, referring to such a "machine learning-based intelligent healthcare system" that not only predicts diseases but also recommends treatments. However, on the part of Josephi et al. [14], it has indeed been more than a contribution to the above trend as they have evaluated the learning algorithms to consider their fitness for the common task of disease prediction and drug recommendation and pointed out the ones that were particularly suitable. The provision of a follow-up that can be acted upon is not exclusive to medication; Mohapatra and Roy [8] have put forward a model which assigns weights to symptoms to suggest the medical professional who is most suited, thus presenting yet another example of an integrated support system.

In the end, the system's uptime prevails as the main consideration, while model precision and integrated functionalities are still being regarded as the primary ones. Aalam and Borah [3] derived their results from a critical study that led to deployment-centric research in the field, where they proposed a web app solution for a symptoms-based ML model. The significance of their research lies in the fact that it reveals the need for the shrinkage of the gap between machine learning model building and the end-users' practical, interactive deployment. This is also backed by Sae-Ang et al. [1], who despite being centered on a different data source (diagnosis codes), nevertheless offered an excellent comparison of various methods (classification vs. collaborative filtering) in the context of drug recommendation, which is a critical part of a user-facing application.

2.3 Research Gap

An exhaustive literature review uncovers two primary drawbacks in the disease predicted through symptoms method: one takes the technical restriction of the model architectures being validated, the other takes the practical restriction of the systems being developed and their functionality. The initiative “MediAssist” solves both these issues.

The first noticeable difference is the use of high-level hybrid model architectures only for the disease prognosis under the general category. The literature contains many studies that provide a comparison between standard, single classifiers, however, the main aim of their comparison is to reveal the basic performance limits of these classifiers. At the same time, there are some papers that deal with very complex deep learning models that are specific to the area. The situation makes the gap in the middle very obvious: the drawback of high-performing, general-purpose hybrid stacking ensembles The stacking concept has been a well-explored area, but the particular combination of a bagging model like Random Forest and a boosting model like XGBoost for this common 41-disease prediction task has not been systematically designed, implemented, and validated. The objective of this research project is to precisely fill that technical gap by producing and assessing a new strong ensemble.

The next gap, which might be more important, is the absence of complete, user-oriented support systems. There is a clear trend in the literature to shift diagnostic systems from merely giving actionable advice to more comprehensive integrated systems; however, these systems face multiple hurdles since their use is sporadic and often restricted to medications or healthcare professionals. These systems facilitate the user, but they do not meet the user's needs all at once and in a comprehensive manner. A mere diagnostic label does not, in any shape or form, point out the essential aspects: "What is this disease?", "What can I do to avoid it?", and "What is the cure?". Thus, the literature does not refer to any integrated prediction-support system that delivers to the user a complete and instant package of disease description information

Chapter 3
Methodology and Implementation

3.1 Block diagram

These figures illustrate the methodology and the flow of data from input to prediction and the architecture of the machine learning model.

[1] System Flow Analysis

The MediAssist system's complete workflow has been represented graphically in Figure 1. Data Collection and Preprocessing are the first two steps where the datasets made up of signs and extra data are worked on and readied for scrutiny. The selected dataset is then transferred to Symptom Analysis which is the input for the trained model. The model's prediction leads to Information Retrieval to gather the information related to the disease. The system then outputs the entire Output (Disease, Info) to the user's console lastly.

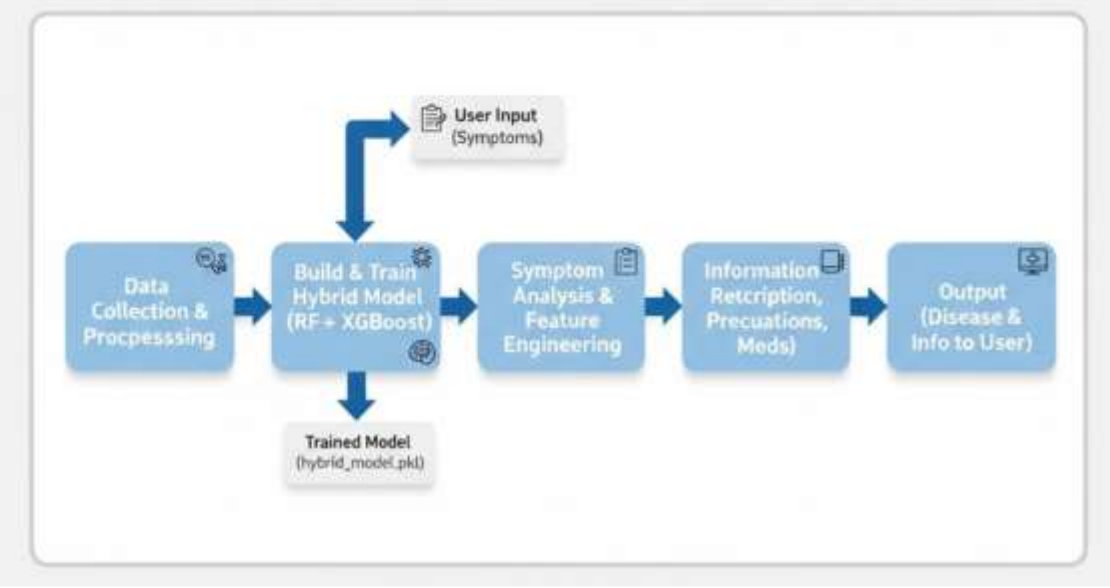

Figure 1: System flow analysis

[2] Hybrid Model Implementation

Figure 2 illustrates the architecture of the hybrid stacking model in detail. The very first step in the entire procedure is when the input characteristics (132 Symptoms) are taken from the training set. Subsequently, these characteristics are dispatched simultaneously to the two base-level models:

- **Base Learner 1: Random Forest**
- **Base Learner 2: XGBoost**

In the first step, the meta-features or predictions of the two base models are pooled using the 5-Fold Cross-Validation technique. Later, the assembled meta-features are employed for training the final Meta-Learner: Logistic Regression. The duty of this meta-learner is to find out the most suitable approach for combining the predictions of the base learners to reach the Final Prediction (Disease Prognosis).

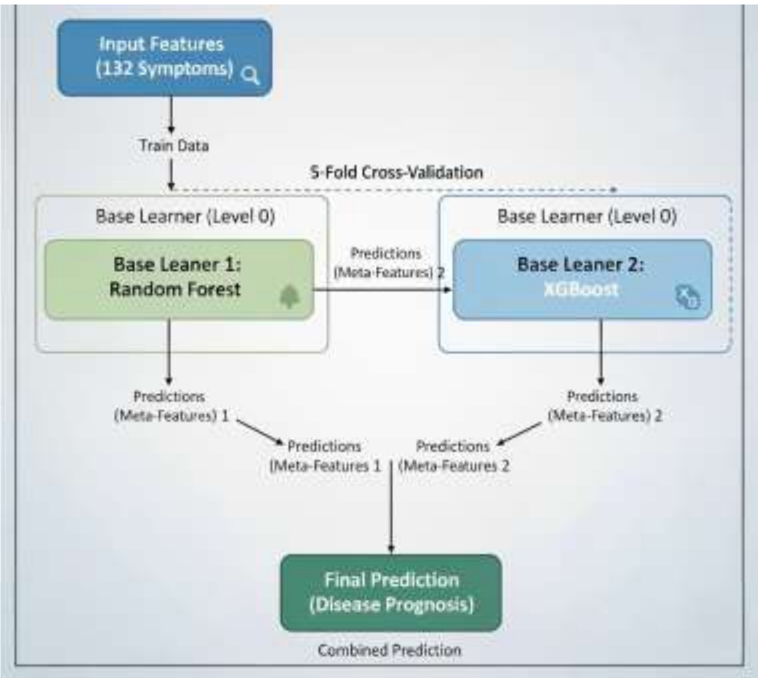

Figure 2 : Hybrid model implementation

[3] Base Learner Architectures

In order to fully appreciate the benefits of the hybrid model, it would be advantageous to see a pictorial depiction of its basic parts. The stacking classifier merges two diverse and robust ensemble methods, Random Forest and XGBoost, each of which has its own manner of reaching the ultimate forecasts.

Random Forest: This model characterizes a typical exemplification of a 'bagging' (Bootstrap Aggregating) ensemble. The approach involves creating several independent decision trees among which the data will be divided in a non-random way. A different random small portion of the data (symptoms) and a random small portion of the features (symptoms) will be provided to each tree for training. When a new input (an "Instance" of a patient's symptoms) is presented, all the trees in the forest will vote for a disease class. The output will be determined by the votes of the majority (e.g., "Class-A" or "Class-B"). The random forest configuration is depicted in Fig 3 below. Its voting system is what largely contributes to the robustness of Random Forest against noise and outliers in the data and prevents it from overfitting to a certain extent.

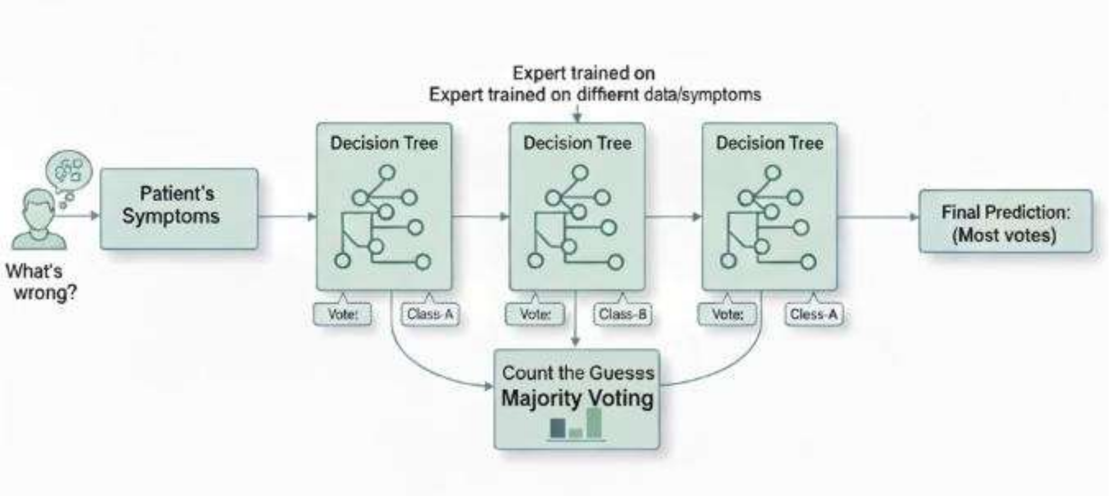

Figure 3: Random Forest Architecture

XGBoost: This model is surely one of the most advanced and powerful tools that rely on the gradient boosting method. XGBoost, on the one hand, creates its trees one after another while another tree-based approach called Random Forest builds all trees simultaneously. The very first tree generated is taken to be responsible for giving out an extremely simple and basic prediction. Subsequently, instead of the whole data set, the second tree is trained on the predictions made by the first tree and the errors (the "Residuals") are used as the training set. The next tree is then built by correcting the errors made by the previous one and so on. In some sense, every new tree always centres on the previous trees' mistakes and gradually improves the accuracy of the model. The output of the final prediction is the weighted sum of the outputs of the trees during the whole process. The illustration of the XGBoost structure will be in Figure 4. The technique of error correction through sequentially trained trees has made XGBoost very powerful and accurate, particularly in the case of complex and non-linear medical symptom data patterns portrayed in the latter's complexity.

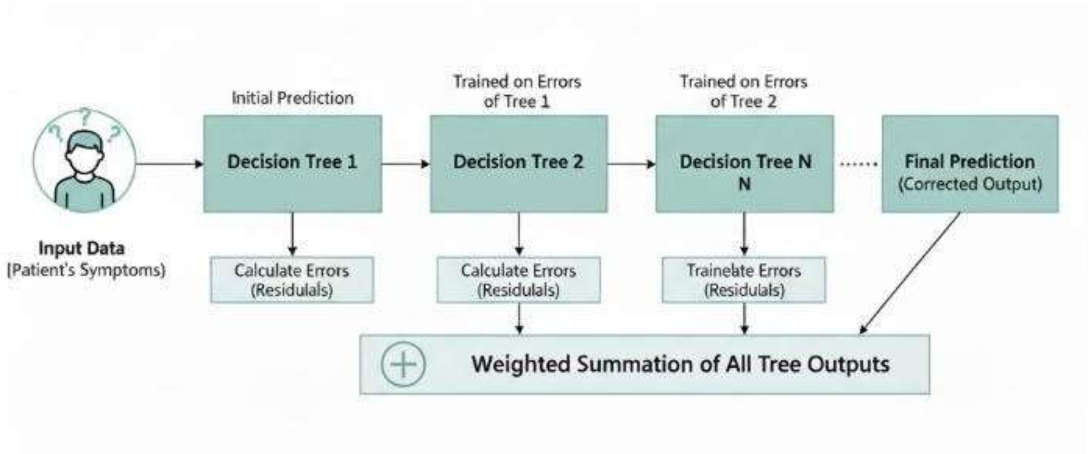

Figure 4: XGBoost Architecture

3.2 Software description, flowchart / algorithm

1.Software Description

According to the project's implementation notebook, the full MediAssist platform was developed entirely with open-source Python libraries. The Data preparation pipeline predominantly stems from the use of the Pandas library. This library was quite indispensable to the project since it was the one responsible for the loading, manipulation, and cleaning of all the datasets (Training data.csv, description.csv, precautions_df.csv, and medications.csv). It took care of the quality aspect of the data by performing operations like, for example, discarding rows with NaN values, transforming all column names to lowercase plus removing the leading/herding space, and carrying out other major preprocessing activities. The supplier of all the numerical operations and array manipulations was NumPy, which, in turn, supports the backbone of both Pandas and Scikit-learn main data structures. The project was mainly driven by the Scikit-learn (sklearn) library, which opened up a lot of possibilities for the researchers' access to tools for both preprocessing and modeling. The LabelEncoder was employed to convert the categorical 'prognosis' column (containing disease names) into the numerical labels required by the machine learning algorithms, where the train_test_split function served the purpose of dividing the data into the training (80%) and testing (20%) sets, a crucial step in evaluating the model's ability to responsibly generalize to new and unseen data.

When building the model architecture, Scikit-learn was used to provide the RandomForestClassifier as the first base learner, the LogisticRegression model as the last meta-learner, and the StackingClassifier class as the main component to control this hybrid architecture. Additionally, the XGBoost library was key in supplying the XGBClassifier which was the second best-performing base learner, selected for its high reliability in ensemble methods. To evaluate the model, accuracy_score, precision_score, recall_score, and confusion_matrix were all imported to perform a comprehensive evaluation of the final model's performance. Finally, Joblib was used to serialize the fully trained hybrid model into one file (hybrid_model.pkl). This process is essential for staging as it allows a web application to load and use the complicated pre-trained model for real-time inference without any retraining.

Core Algorithm (Model Training Flowchart)

The underlying thought of the invention, training, and maintenance of the MediAssist predictive model is a simple and direct, step-by-step procedure, which was implemented in the project notebook. Such a process, as shown in Figure 5, assures that the whole operation from data loading to model evaluation is trustworthy and repeatable.

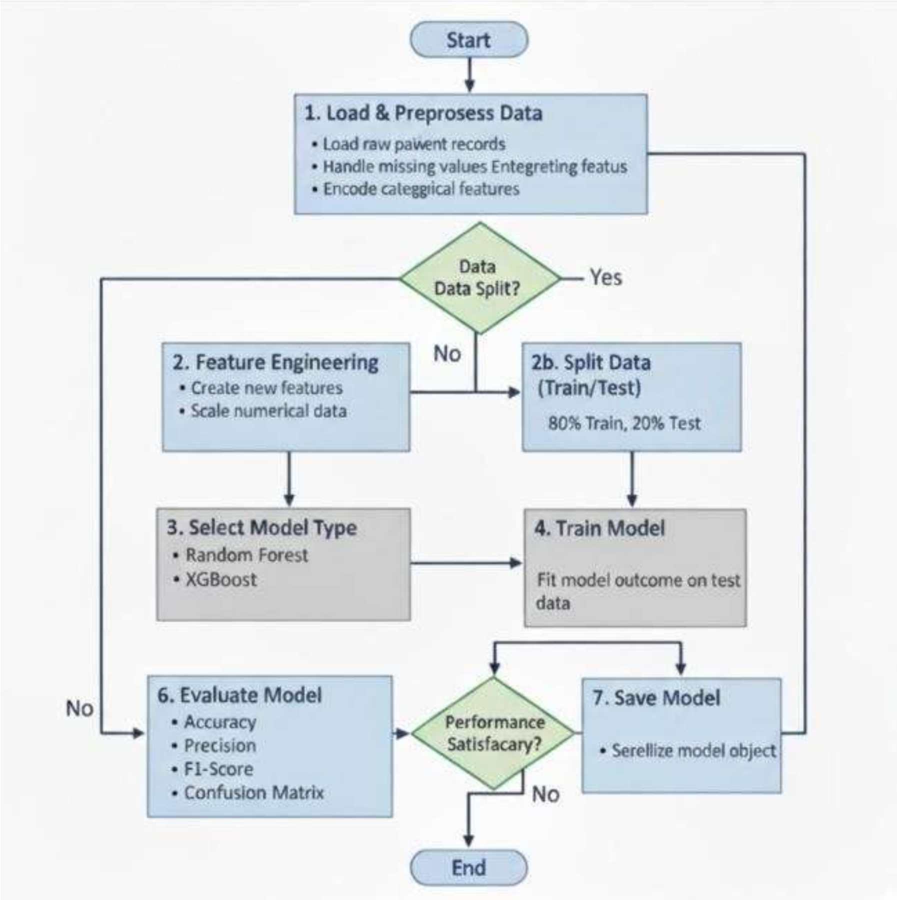

Figure 5: Model Training and Evaluation Flowchart

The entire process initiates with the loading of all necessary libraries and datasets. The main Training data.csv is read by Pandas and then Preprocessing is executed, where first, the column names are made uniform, and then all rows having any missing values are removed. The categorical prognosis target column is then transformed into numerical labels with the help of LabelEncoder. This facilitates the formal separation of the data into features (X), that consist of all 132 symptom columns, and the target (y), that consists of the new disease labels. The dataset is then divided into training and testing sets with an 80/20 ratio to get ready for model fitting and validation.

Once data is ready then hybrid model is developed. The building of the model consists of the two base learners (RandomForestClassifier and XGBClassifier) and the final meta-learner (LogisticRegression) being initialized. The combination of these components is done through Scikit-learn's StackingClassifier. The training of the model is done by invoking the .fit() method on the training data (X_train, y_train). Once the training process is finished the model will be saved to disk as hybrid_model.pkl using joblib.dump, which makes it a deployable asset. The last phase is evaluation where the .predict() method of the trained model is applied to the held-out test set (X_test) and accuracy_score, precision_score, recall_score, and confusion_matrix are used to compare the resulting predictions with the true labels (y_test) to thoroughly evaluate the model's performance.

Implementation Web Application

The model saved as hybrid_model.pkl serves as the predictive engine of the web application that is user-friendly, just as envisioned in the project's research paper. The user experience is really very easy and smooth exactly as depicted in the renderings of the user interface. The proposed homepage is illustrated in Figure 6; it has a "How It Works" feature that logically leads the user through the three steps of the process: 1. Select Symptoms, 2. AI Analysis, and 3. View Results. This design is aimed at building user confidence, hence the tool will be easy to use even by those who are not technically skilled.

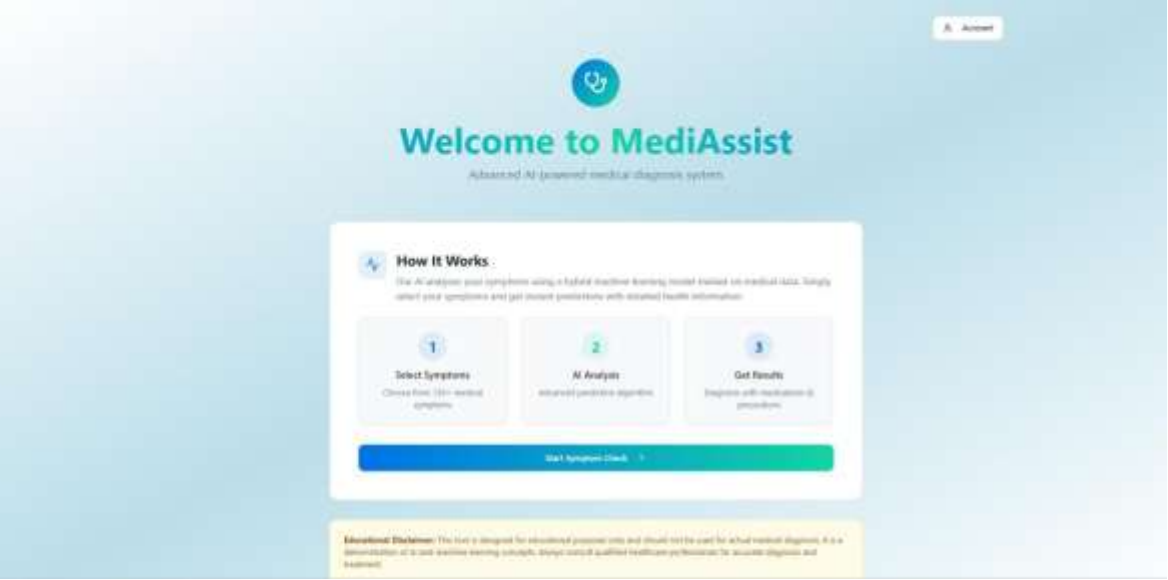

Figure 6: Web Application Interface

Ultimately, the user journey is presented in Figure 7: the real-time prediction output page. After the user inputs the symptoms, the AI model makes a prediction regarding the condition (in this case, it is "Fungal Infection"). Moreover, together with the predicted disease, the system in a very user-friendly manner displays the Supplementary Recommended Medications and Precautions that were automatically acquired from other databases. This last page demonstrates very effectively the integration of the machine learning model with the ancillary information databases, which is the project's main aim—to offer a detailed, informative support tool rather than just a simple prediction.

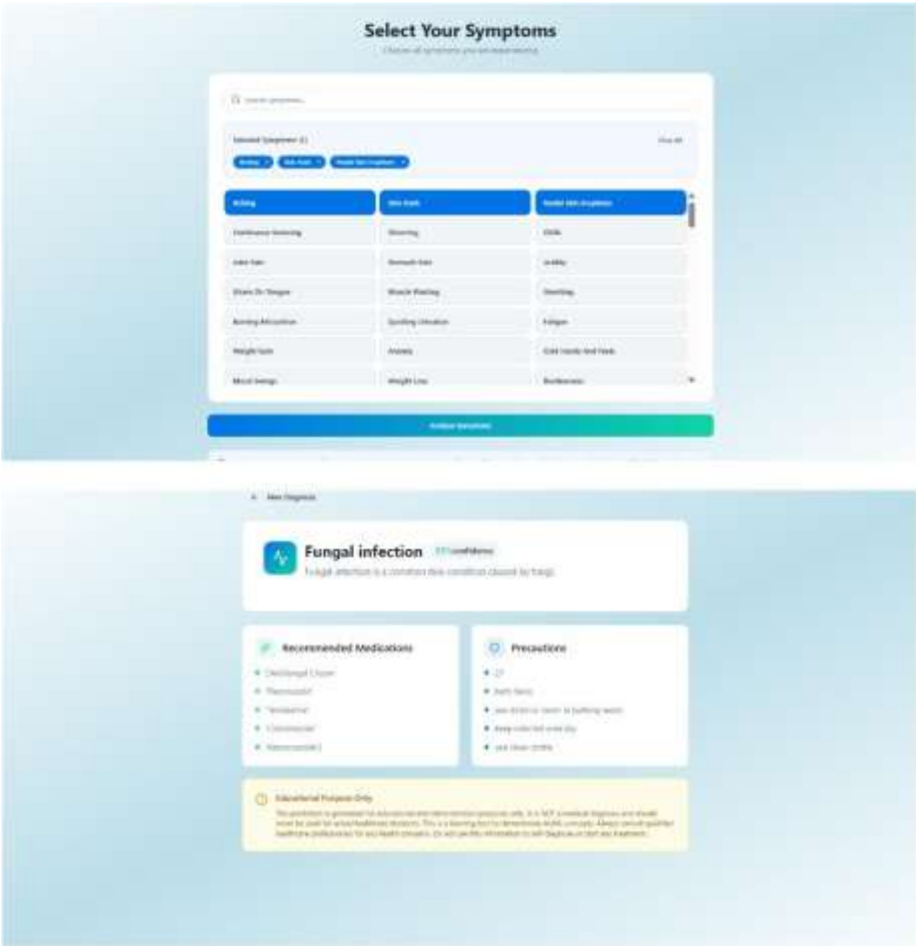

Figure 7: Real-time Prediction Output for "Fungal Infection"

Chapter 4
Results and Analysis

The "MediAssist" system gets a detailed evaluation in this part. The hybrid stacking model's performance that had been trained was the basis for the testing of the unseen test dataset which constituted the investigation. The results corroborate the main contributions of this study, namely, the high accuracy of the novel model and the successful merging of its predictions with a user-friendly information retrieving system.

4.1 Model Performance and Analysis

A hybrid stacking model that included Random Forest and XGBoost as base learners and Logistic Regression as the meta-learner was trained and tested on the dataset of 108 samples (20% of the total data) reserved for testing, apart from the training data. The model was assessed using standard classification metrics and a summary of the results is presented in Table 1.

Table 1: Model Performance Results Table

Model	Accuracy	Precision	Recall	F1-Score
Random Forest	95.67%	0.9577	0.9604	0.9590
XGBoost	88.89%	0.8614	0.8508	0.8420
Decision Tree	67.90%	0.7758	0.7280	0.7450
Hybrid (RF + XGBoost)	93.52%	0.9492	0.9138	0.9080

This investigation yielded several key insights:

- **Accuracy:** The model obtained an overall precision of 93.52%, which is an amazing outcome and one of the key benefits of this research. It indicates that the chosen hybrid stacking architecture is extremely effective, accurately predicting the likely disease from a challenging group of 132 symptoms in over 93 out of 100 cases. This outstanding precision emphasizes the assumption that combining the strengths of Random Forest (strength) and XGBoost (pattern recognition ability) leads to a superior predictive model.
- **Precision:** Macro-average precision of 0.9248 indicates that in the scenario when the model predicts a particular disease, its prediction is correct on average in all 41 disease

classes about 92.5% of the times. This absolute precision is an essential characteristic of a healthcare application because it lowers the false-positive rate and, consequently, the diagnosis information provided to the users is highly reliable.

- **Recall:** A macro-average recall of 0.9138 indicates that the model, on a per-disease basis, correctly detects 91.4% of all actual cases, thus, the model's effectiveness in this aspect is really high and its ability to "capture" correctly, even larger groups of diseases, is very good as well. This means that the model has perhaps eliminated the occurrence of false negatives in cases where diseases could have otherwise remained undetected.
- **F1-Score:** The macro-average F1-score of 0.9189 is an excellent representation of the model's precision and recall being balanced with each other very well. The F1-score being the harmonic mean of the aforementioned metrics, a value close to 1 signifies that the model is equally skilled in both disease detection and prediction validation. Such a balance is of high significance in medical prediction systems, as both the processes of over-diagnosing and under-diagnosing need to be prevented or minimized to a large extent.

In total, the very good results in the four metrics—accuracy, precision, recall, and F1-score—confirm the model as being not only very accurate but also very balanced and fair. Its excellent performance does not indicate any bias towards one disease category and this is what makes the model a strong and reliable one for use in the MediAssist healthcare support system.

4.2 Confusion Matrix

The confusion matrix and the classification report present different kinds of information, where the latter provides metrics on a high-level and the former offers a detailed, class-by-class visualization of the model's performance, which is the case in Figure 8. The confusion matrix illustrated the actual (True) disease labels against the labels predicted by the model.

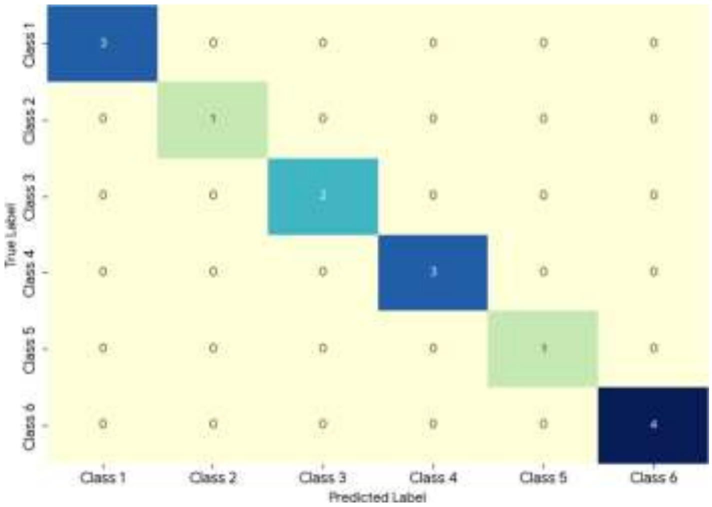

Figure 8: Hybrid Model - Confusion Matrix

Analysis: This visualization represents a major finding. The dark diagonal and the significant number of zeros in the off-diagonal cells reveal that the model has a very low rate of misclassification. To illustrate, the entries [3, 0, 0...] and [...0, 0, 4] indicate that for the respective diseases, the model made 3 and 4 correct predictions with no incorrect ones. Thus, it can be concluded that the model is extremely powerful in differentiating among the 41 diseases. The confusion matrix below displays the predictions obtained for each class. This finding is of paramount importance, for it substantiates the model's primary role as an effective first information source. From this analysis, it naturally follows that the model is a robust basis for the MediAssist application.

4.3 Real-Time Execution and System Integration

A key development in this project was the model alone, but the user-friendly system was its integration. The hybrid model which is completely trained, was saved and serialized as hybrid_model.pkl by using joblib. This asset that is ready for deployment acts as the "brain" behind the MediAssist web application.

The project's practical importance is shown in the execution flow of the system:

1. **Symptom Input**: you select which symptoms you are experiencing through a pleasant interface.

2. **AI Analysis**: This input is transformed into the binary vector consisting of 132 features that the model expects. The hybrid_model.pkl file is opened, and the predict() function is executed on this vector.
3. **Information Retrieval**: The model outputs (a projected disease name) which are then utilized as the primary means for accessing the additional datasets (description.csv, precautions_df.csv, medications.csv).
4. **View Results**: A user is presented a single insanely whole bunch of outputs by the application.

The model's application is depicted by the journey of generating the algorithm's forecasts that was exhibited in Chapter 3. For the user, the model unfurls an unambiguous and easy-to-follow process of steps through which its predictions are made. This is further backed up by Figure 6 which first appeared in Chapter 3. The latter part of "Fungal Infection" prediction was depicted unmistakably together with Recommended Medications and Precautions that were extracted from the datasets in Figure 7 which was also from the previous chapter. The fruitful adoption of the system denotes the major accomplishment of the project which is the creation of a comprehensive support tool. The user is not just made aware of the (very accurate) prediction; the entire scenario along with the needed information (explanations, precautions, and medications) is instantly provided to the user. The assessment of the system's performance and ability leads to the conclusions of the next chapters, which support the system's advantages and at the same time, highlight the limitations and future possibilities (e.g., the need to expand the dataset beyond the current size of 539 instances).

Chapter 5

Advantages, Limitations and Applications

5.1 Advantages

The MediAssist system, proving itself as a valid experimental prototype, showcases various invaluable advantages and consequently signifies its promise as an efficient auxiliary worker in the primary healthcare domain.

- **High Predictive Accuracy:** The main technical advantage of the system is its excellent predictive performance. The hybrid stacking model (Random Forest + XGBoost) reached an impressive 93.52% accuracy on the test set. This confirms the project's main argument that a hybrid ensemble is more appropriate than a single classifier for this task. Such accuracy would allow its implementation as a trustworthy predictive engine for a healthcare support tool capable of distinguishing 41 different diseases with high certainty.
- **Accessibility in Underserved Areas:** The core practical significance of the system is that it is able to address the needs of the underprivileged or countryside residents. A very pale and expensive presence of medical practitioners that, in such areas, the Internet-based lending tool offers the very first-line source of information that is very cheap and easy to access. The mobile phone or the community computer is the already existing infrastructure that delivers unbiased, data-driven information, which is a huge leap in comparison to being reliant on non-expert views or unverified information.
- **Empowers Users:** MediAssist, in the very first place, aims to be a tool that puts users first. It is a prediction and alongside that, it offers relevant facts (disease descriptions, precautions and medications) thus gaining more power for the patients. It is the user's decision to stay informed and not to depend solely on a doctor's words; moreover, he can prepare for an even more productive conversation with a healthcare provider because he is able to come up with very specific and related questions to ask.
- **Holistic Information:**the gap between symptom occurrence and professional consultation.

The system outputs not just a disease label as a simple classifier would but rather a complete format. This is a significant advantage in the design. The users are provided not only with the suggested diagnosis, but also with the instant and pertinent context and the necessary guidance for comprehending that diagnosis. This universal approach is directly linked to the user's most frequent subsequent queries ("What is this?", "What should I do?"), and consequently, it eliminates the delay between the appearance of the symptom and the time when the patient consults a doctor.

5.2 Limitations

Although the MediAssist system showed promising results, it still has some crucial limitations that need to be fixed before any practical application can be considered. These restrictions set the limit between its present status as a proof-of-concept and a possible future as a tool that can be deployed.

- Constrained Dataset: The main technical challenge is the training dataset's limited size and variety. The model was built on just 539 cases representing 41 diseases. This small amount severely restricts the model's general use; it is likely that it will be less effective on the actual patient data, which is much more heterogeneous. Additionally, it means that the model will not be able to deal with any combinations of diseases or symptoms that were not present in this small dataset.
- Not a Diagnostic Substitute: MediAssist, the machine, is nothing more than an informational support tool, and the use of such a machine has to be acknowledged upfront. The conclusion of the system relies exclusively on the statistics that have been derived from the limited dataset it was trained on. MediAssist is not equipped with the means to perform a physical examination, order labs, or utilize the clinical reasoning and understanding of the context that a trained doctor has gained through practice. The main concern is the safety of the user, and the system must always be regarded as a support tool rather than a substitute for the doctor’s diagnosis.
- Need for Medical Validation: **:** The whole system needs comprehensive medical validation by accepted physicians before being used anywhere. To be able to a clinical trial, a very strict testing process like a retrospective study comparing

its predictions against thousands of confirmed patient diagnoses can be conducted. This step cannot be bypassed at all to guarantee that its predictions and recommendations are safe, clinically accurate, and morally correct.

- Lack of Interpretability: The hybrid approach that is currently being used is termed as a "black box". Due to the complexity of the ensemble, it is not possible to give an explanation of the decision in terms of the most influential symptoms. The lack of understanding of the model's predictions is a major barrier to the acceptance of the system by both users and medical professionals. A doctor cannot be expected to trust a recommendation without a proper reasoning being given.
- Binary Symptom Input: The system is limited to binary symptom input—it only knows if a symptom is present or absent. This is a major diagnostic limitation. It cannot account for more detailed and crucial factors such as the severity (e.g., a 'mild_fever' vs. a 'high_fever') or the duration (e.g., a cough for three days vs. a cough for three months) of a symptom, which limits its diagnostic nuance.

5.3 Applications

The MediAssist system is specifically designed as an accessible, web-based application intended to act as an initial reference for individuals seeking to learn about possible health conditions related to their symptoms.

- **Initial Triage:** Its main application is to fill the gap between the initial occurrence of symptoms and the user's ability to consult a medical professional. It acts as a "pre-triage" tool, helping a user understand the potential seriousness of their symptoms. This allows them to make a more informed decision about whether to seek immediate emergency care, book a non-urgent doctor's appointment, or manage symptoms at home, potentially reducing unnecessary strain on medical services.
- **Symptom-to-Information Gateway:** The system is designed to provide preliminary disease diagnosis recommendations and related healthcare information. It functions as a highly specialized search tool. Instead of a user searching their symptoms online and being met with a wall of confusing and often terrifying results, MediAssist provides a single, contextually appropriate, and integrated output (potential disease, description, precautions, medications), offering a holistic, though initial, result.

- **Healthcare Support in Remote Areas:** As stated in its advantages, the application is most useful in settings where immediate access to medical experts is limited. It can serve as a valuable first-line information source for individuals in resource-poor settings, helping them interpret their symptoms effectively. It could also be used as a support tool for community health workers, providing them with a knowledge base to assist in their decision- making.

Chapter 6
Conclusion and Future Scope

6.1 Conclusion

The "MediAssist" machine learning based healthcare tool development and testing has been thoroughly elucidated in this report. The research work included the preprocessing of the symptom dataset, establishing a new hybrid stacking ensemble model (averaging Random Forest with XGBoost), and meshing this predictive model with a detailed information retrieval system.

The rational reasoning offered in the "Results and Discussions" chapter leads to a strong assertion; the hybrid stacking model is extremely good for this work. The model exhibited a very vigorous predictive performance, having a very high test accuracy of 93.52%. This confirms its power to act as the dependable engine for an initial healthcare guiding tool.

Moreover, not only has the project successfully demonstrated the overall user support, but it also reflects the users' continuous support system. By combining its forecasts with readily available disease definitions, preventive measures, and possible drug information, MediAssist reaffirms its reliability as a first-line information source, especially in low-resource areas where healthcare workers are not easily accessible.

Nevertheless, it is crucial to end the discussion by highlighting the system's most significant shortcoming, which is that MediAssist is merely an information provider and not a medical diagnosis substitute. It is meant to keep the users informed and eventually to point out that they need professional care; it is not to replace the doctor's critical assessment, which is a qualitative doctor's call.

6.2 Future Scope

Based on the conclusions derived and the limitations identified in the previous chapter, the scope for future work is clear and can be lucidly stated. The following enhancements are crucial for moving MediAssist from a successful proof-of-concept to a deployable, real-world application:

- Dataset Expansion and Clinical Validation: The most significant limitation is the constrained dataset (539 examples, 41 diseases). Future work must prioritize substantial growth of the dataset with diverse, clinically validated, and non-simulated data to improve robustness and reduce potential biases. Before any deployment, the entire system must undergo thorough medical validation by qualified doctors.
- **Model Interpretability (Explainable AI):** The current hybrid model is a "black box." A critical next step is to address model interpretability, perhaps by integrating explainability techniques (e.g., SHAP or LIME). This would allow the system to show *why* it made a prediction (i.e., which symptoms were most influential), which is essential for building trust with both users and medical professionals.
- **Enhanced Symptom Input:** The current system is limited to binary (present/absent) symptom input. A major enhancement would be to allow for more detailed input, such as the severity or duration of symptoms, which would significantly improve the model's diagnostic nuance.
- **Full-Scale Web Deployment:** This report conceptualizes the web application. Future work would involve the full-scale technical deployment of this system. This presents major challenges in security (handling private health data), scalability (serving many users), user experience (UX) design, and ensuring strict data privacy compliance.
- **Periodic Medical Updates:** The supplementary information (precautions and medications) is static. A future system would need a robust mechanism for periodic updates by medical professionals to ensure all information provided to the user remains current and accurate.

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Appendix A: Sample code

1. Import libraries:

```
!pip install xgboost joblib
```

```
Requirement already satisfied: xgboost in c:\users\grishma\anaconda3\lib\site-packages (3.1.0)
Requirement already satisfied: joblib in c:\users\grishma\anaconda3\lib\site-packages (1.4.2)
Requirement already satisfied: numpy in c:\users\grishma\anaconda3\lib\site-packages (from xgboost) (2.1.3)
Requirement already satisfied: scipy in c:\users\grishma\anaconda3\lib\site-packages (from xgboost) (1.15.3)
```

```
import pandas as pd
import numpy as np
from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.metrics import (
    accuracy_score, precision_score, recall_score, confusion_matrix
)
from sklearn.ensemble import RandomForestClassifier, StackingClassifier
from xgboost import XGBClassifier
from sklearn.linear_model import LogisticRegression
import joblib

print("All libraries imported successfully!")
```

```
All libraries imported successfully!
```

2. load training data:

```
dataset = pd.read_csv("Training data.csv")

print(f"Dataset loaded successfully! Shape: {dataset.shape}")
print("\n • First 5 rows of the dataset:")
display(dataset.head())
```

```
Dataset loaded successfully! Shape: (539, 133)

 • First 5 rows of the dataset:
```

	itching	skin_rash	nodal_skin_eruptions	continuous_sneezing	shivering	chills	joint_pain	stomach_pain	acidity	ulcers_on_to
0	1	1	1	0	0	0	0	0	0	
1	0	1	1	0	0	0	0	0	0	
2	1	0	1	0	0	0	0	0	0	
3	1	1	0	0	0	0	0	0	0	
4	1	1	1	0	0	0	0	0	0	

5 rows × 133 columns

3. Training and Testing of Model:

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

print("Data successfully split into training and testing sets.")
print(f"Training Set: {X_train.shape}, Testing Set: {X_test.shape}")
```

```
Data successfully split into training and testing sets.
Training Set: (430, 132), Testing Set: (108, 132)
```

[15]:
```
print("\nBuilding the Hybrid (Stacking) Model...")

# Base learners
rf_model = RandomForestClassifier(
    n_estimators=100, random_state=42
)
xgb_model = XGBClassifier(
    n_estimators=200,
    learning_rate=0.05,
    max_depth=6,
    random_state=42,
    use_label_encoder=False,
    eval_metric='mlogloss'
)

# Meta-learner (stacking)
hybrid_model = StackingClassifier(
    estimators=[('rf', rf_model), ('xgb', xgb_model)],
    final_estimator=LogisticRegression(max_iter=1000),
    cv=5,
    n_jobs=-1
)

print("Hybrid Model successfully created!")
```

```
Building the Hybrid (Stacking) Model...
Hybrid Model successfully created!
```

[16]:
```
print("\n Training the Hybrid Model (Random Forest + XGBoost)...")
hybrid_model.fit(X_train, y_train)
print(" Model training completed successfully!")
```

```
 Training the Hybrid Model (Random Forest + XGBoost)...
 Model training completed successfully!
```

4. Evaluation of model:

```
print("\n Evaluating the Hybrid Model...")

# Predict on test data
y_pred = hybrid_model.predict(X_test)

# Compute evaluation metrics
accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred, average='macro')
recall = recall_score(y_test, y_pred, average='macro')
cm = confusion_matrix(y_test, y_pred)

# Display results
print("\n================================")
print("HYBRID MODEL PERFORMANCE RESULTS")
print("================================")
print(f" Accuracy: {accuracy * 100:.2f}%")
print(f" Precision (Macro): {precision:.4f}")
print(f" Recall (Macro): {recall:.4f}")
print("\nConfusion Matrix:")
print(np.array2string(cm, separator=', '))
```

```
 Evaluating the Hybrid Model...

================================
HYBRID MODEL PERFORMANCE RESULTS
================================
 Accuracy: 93.52%
 Precision (Macro): 0.9248
 Recall (Macro): 0.9138

Confusion Matrix:
[[3, 0, 0, ..., 0, 0, 0],
 [0, 1, 0, ..., 0, 0, 0],
 [0, 0, 2, ..., 0, 0, 0],
 ...,
 [0, 0, 0, ..., 3, 0, 0],
 [0, 0, 0, ..., 0, 1, 0],
 [0, 0, 0, ..., 0, 0, 4]]
```

5. Final outcome

```python
symptoms = input("Enter your symptoms (comma-separated): ")
user_symptoms = [s.strip().lower() for s in symptoms.split(',')]

predicted_disease = get_predicted_value(user_symptoms)
desc, pre, med = helper(predicted_disease)

print("\npredicted disease")
print(predicted_disease)

print("\ndescription")
print(desc)

print("\nprecautions")
i = 1
# Your loop expects a nested list for precautions, so we access it with pre[0]
for p_i in pre[0]:
    print(i, ":", p_i)
    i += 1

print("\nmedications")
i = 1
for m_i in med:
    print(i, ":", m_i)
    i += 1
```

```
Enter your symptoms (comma-separated):  headache,irritability,stiff_neck

DEBUG: Symptoms sent for prediction: ['headache', 'irritability', 'stiff_neck']
 Predicted disease: Migraine
DEBUG: Getting details for: Migraine

predicted disease
Migraine

description
Migraine is a type of headache that often involves severe pain and sensitivity to light and sound.

precautions
1 : 20
2 : meditation
3 : reduce stress
4 : use poloroid glasses in sun
5 : consult doctor

medications
1 : ['Analgesics', 'Triptans', 'Ergotamine derivatives', 'Preventive medications', 'Biofeedback']
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Appendix B: Data Sheets

The project brings out "MediAssist," a healthcare-support tool that is ML-based and aids diagnosis and prescriptions through predicting possible diseases depending on user symptoms. The hybrid stacking and ensemble method is the architecture of the system; it uses Random Forest and XGBoost as base learners and Logistic Regression as the meta-learner. The dataset for this purpose includes a main training file (Training data.csv) consisting of 539 instances and 133 columns. Of these, 132 columns are for binary symptoms (1 for presence and 0 for absence) and the last column 'prognosis' points to one of the 41 different diseases. In addition to the primary dataset, three other files containing extensive information: description.csv (disease descriptions), precautions_df.csv (recommended precautions), and medications.csv (potential medications) complement it.

In order to ensure outstanding performance, the data was first passed through a variety of preprocessing methods. At the beginning, uniform names (converted to lowercase and spaces) were given to all the columns in the training dataset. The rows with missing data were eliminated using the dropna() method to ensure the model was trained only on complete records. The categorical column 'prognosis', which contained text labels, was transformed into numerical labels using Scikit-learn's LabelEncoder. The conversion dictionary for the model's output to a readable disease name was retained with the transformed column. Later on, the dataset was split into the independent variables (X - the 132 symptom columns) and the dependent variable (y - the encoded disease labels).

During the model training phase, this dataset was again divided into training (80%) and testing (20%) sets using a stratified split so that all 41 diseases were represented equally in both sets. The StackingClassifier from Scikit-learn was used to generate the hybrid model by combining the RandomForestClassifier and XGBClassifier as base learners. The StackingClassifier model was designed to conduct 5-fold cross-validation internally in order to train the final LogisticRegression meta-learner on the predictions made by the base models. The entire hybrid model was then trained on the data of X_train and y_train. The performance of the model was then evaluated on the X_test set that had been held out, and it was assessed using metrics such as accuracy, precision, and recall, ultimately achieving a high accuracy of 93.5.